

AAYUSH PANDA

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TECHNICAL SKILLS

Programming Languages: C, C++, Rust, Python, Java, TypeScript, SQL, Haskell, Scheme, Bash, HTML/CSS, x86 assembly
Data Science & Machine Learning: NumPy, Pandas, sklearn, PyTorch, TensorFlow, Keras, OpenCV, Matplotlib, Seaborn, Jupyter
Databases: SQL dbs, Firestore, Neo4J, Chroma

EDUCATION

University of Waterloo

Bachelor of Computer Science, Honours Co-op

GPA: 3.92/4.00

Sep. 2024 – Apr. 2028

EXPERIENCE

ML & NLP Engineering Intern

Sep. 2025 – Present

Huawei Canada

- Developed information theoretic algorithm for textual semantic chunking using PMI
- Developed algorithms for KG generation from natural language documents, with approximate named entity linking
- Developing edit localisation algorithms for LLM-powered text editing

Laboratory Assistant

Apr. 2025 – Dec. 2025

University of Waterloo Multi-Sensory Brain and Cognition Lab (MBC)

- Developed CV pipelines and algorithms for baseball/eye tracking for an ongoing research project on how batsmen's eyes track baseballs through various jumps and saccades
- Developed synthetic data pipelines to train a U-net model to detect baseballs in captured video
- Developed correspondence algorithm to synchronise data from high speed head mounted camera and eye tracking data

Undergraduate Research Assistant

Apr. 2025 – Aug. 2025

University of Waterloo Data Privacy Lab

- Research on private record linkage protocols using locality sensitive hashing, supervised by Dr. Florian Kerschbaum
- Developed a differentially private method of estimating max LSH bin, to allow for private frequency smoothing without sharing bin size info between parties (which leaks information)
- Developed k-d cuckoo hashing algorithm for private set intersection using the backing array

Software Engineering Intern

Mar. 2025 – Apr. 2025

Toma (YC W25)

- Wrote a service that evaluates dealership customer support quality using a conversational voice AI built with ElevenLabs. This data was then used to find ideal clients, and demonstrate/evaluate Toma's impact on dealerships.
- Developed a server log viewer with IDE-like features to improve debugging efficiency.
- Worked on model evaluation for Toma's voice agents

PROJECTS

🔗 Concord — Rust, Vulkan, CUDA, HLSL

- Ongoing project to build a cross-platform runtime in Rust for accelerated computation, with a unified ISA and memory model for CPU/GPU operations.

🔗 Semantify — Python, React, FastAPI, Sentence Transformers, UMAP, Ollama

- Built a system that semantically organises **1000+ files** into a directory structure in minutes using embedding models and custom hierarchical clustering and subtree merging algorithms
- Developed an interactive visualization of document embeddings to allow intuitive semantic exploration
- Implemented a RAG-powered chat interface to intelligently query uploaded documents with source citation

🔗 Phased Array SONAR — C++, Python, Xtensa LX6 microprocessor, AVR RISC processors

- Engineered a sub-degree precision beam-steering phased SONAR array with real-time radar-style display on oscilloscopes.
- Designed a waveguide to reduce inter-element pitch and suppress grating lobes, enhancing beam directivity.
- Built a phased array simulator suite to visualize beamforming, steering, and focus behaviors in 2D.

PATENTS

CA 3222437: Device for redirection of optical beams using virtual gratings generated by stationary waves.

CA 3119717: Compliant mechanism for operating flight control surfaces of a remotely piloted aircraft.

AWARDS

University of Waterloo President's Research Award

Jane Street Estimathon @ UWaterloo (2024): First place

Hack the North 2022: Winner (out of 829 participants)

PicoCTF 2022: 2nd place in Canada, 14th (top 0.001%) globally

FIRST Innovation Challenge 2021: Semifinalist